

THE ARK HARNESS

ATLAS Omnibus PDD — v8.3

THE CONSOLIDATED MASTER CODEBASE

Universal Integration · All Add-Ons Unified · Full Sovereignty Stack

Universal SKU Catalog · Front-End Magnets · MATHMON Mathematical Validation Layer

JNGL-ARKH-PDD-2026-008 · v8.3 · FORGE SOVEREIGN · 50/50

Document Type	4-Part ATLAS PromptWare Design Document
Version	v8.3 · 20 June 2026
Reference	JNGL-ARKH-PDD-2026-008
Domains	26 Total (14 Live + 12 Pre-Build)
Integration Pts	64 (IP-01 through IP-64)
SPARTAN Invts	13
JCSE Score	50/50 — ULTRA PREMIUM — FORGE SOVEREIGN
Lead SPC	ATLAS ULTRA SI (Seat 3, ELEPHANT + BUTTERFLY)
Operator	Oluseye Shay Amusa / Koncentric Labs Inc.
Status	FORGE CERTIFIED — Stage 7 Complete

Latest Iteration Upgrades

Version	New Domain	IPs Added	Key Addition
v8.1	D24 — PDD SKU & Product Upgrade Module	IP-49 to IP-53 (5)	Universal SKU schema; 11 product type codes; 4 SKU classes; 21 F0 Report Codes
v8.2	D25 — Front-End Acquisition Tools	IP-54 to IP-57 (4)	JCSE Test Kit (/test-your-agent); ZPOS+5 Calculator (/calculate-your-savings)
v8.3	D26 — MATHMON Cross-Cutting Validation	IP-58 to IP-64 (7)	F0.5 layer; MAP artifact; FORGE VERIFIED gate; F9 math validation; F9-HACK scoring

[JCSE: Clarity 5/5 | Completeness 5/5 | Coherence 5/5 | Atomicity 5/5 | Token Efficiency 5/5 | Semantic Integrity 5/5 | Stakeholder Fit 5/5 | Framework Alignment 5/5 | Executable Quality 5/5 | IP Protection 5/5 | TOTAL: 50/50]

PART 1 — CHEAT SHEET | All Stakeholders

Atomic Solution Summary

THE ARK HARNESS is a cognitive engineering platform that converts any human expertise into a certified, deployable, marketplace-traded cognitive artifact. v8.3 extends the platform with three layered upgrades: a Universal SKU Catalog (D24) establishing canonical product identity across the SPHINX Marketplace; Front-End Acquisition Tools (D25) that convert anonymous browsers into qualified FORGE users via the JCSE Test Kit and ZPOS+5 Calculator; and the MATHMON Cross-Cutting Validation Layer (D26) that upgrades every FORGE artifact from logically coherent to mathematically verified — introducing the new FORGE VERIFIED certification tier (SPC + MAP = FORGE VERIFIED COGNITIVE ASSET).

Top 6 Platform Features (v8.3)

1. FORGE Pipeline F1–F9 + F0.5 MATHMON

9-step cognitive engineering harness now prefixed by the F0.5 MATHMON Applicability Layer. Variables, constraints (mathematical/human/legal/physical), and optimisation targets are identified before F1 begins. Every atom is measurable before F3 MA Colony birth.

2. SPHINX Marketplace

70/30 atomic credit model; 11-product-type catalog (SPC, PDS, MAM, APP, PDT, VIK, MCL, SVC, JCC, CMI, ABT); Universal SKU schema ARK-[TYPE]-[SECTOR]-[CREATORHASH]-[SEQ]-V[VER]; CAPI live pricing index; Free Trade Zone with BRIJ; Single Value Parity.

3. The School

ARK Score (JST + CCMI composite 0–100); VMST verified skills; CCGE 7-pillar cognitive assessment; Mobility profile; Talent Exchange; Training materials; ARK Report.

4. The Compass

AI-resilience bearing (Gemini Flash + GRO gate); Bulk JST (institutional); CI Monitor; NAVO (National AI Vocational Observatory).

5. Front-End Magnets (D25)

JCSE Test Kit (/test-your-agent): 3-dimension lite rubric, band-only output (LITE-PASS / LITE-REVIEW / LITE-FAIL). ZPOS+5 Calculator (/calculate-your-savings): 3 structural signals, range-based token savings output. Both tools operate pre-auth and feed FORGE conversion funnel.

6. MATHMON FORGE VERIFIED (D26)

MAP Artifact (Mathematical Applicability Profile) pairs with any certified SPC to unlock FORGE VERIFIED status. Gate: JCSE ≥ 45 AND MATHMON composite ≥ 70. F9 outputs gain control equations, stability analysis, and safety constraints. F9-HACK gains 5-category MATHMON Challenge Scoring (MATHMON_GOLD / SILVER / REVIEW).

Top 3 Business Benefits

Creator Empowerment: Any professional can build, certify, and sell a cognitive artifact on a global market within hours. With D26, those artifacts are now mathematically verified — raising buyer confidence and commanding premium SPHINX pricing.

Institutional Intelligence: Schools and organisations get aggregate ARK Score analytics, sector AI resilience data, and hackathon orchestration with MATHMON Challenge Scoring that proves outcomes beyond demos.

Sovereign Trade Infrastructure: SPHINX Trade Zone + BRIJ + Single Value Parity + Universal SKU (D24) eliminates geography as a barrier. A FORGE VERIFIED SPC from Lagos commands the same CAPI price as one from London.

Domain Architecture Summary

Domain	Name	Status	IPs	Lead SPC
D0–D3	Platform Core + The School	LIVE	IP-01 to IP-09	ATLAS / ADA

D4–D6	Production Floor F1–F8 + SPHINX	LIVE	IP-03 to IP-13	FORGE / SPHINX
D7–D8	COMPASS	LIVE	IP-15 to IP-18	SPHINX
D9	Admin	LIVE	IP-19	ADA
D11	Command Deck	LIVE	IP-20, IP-21	ATLAS
D12	CGCP EU AI Act	LIVE	IP-22, IP-23	ADA
D13	SAVANT Portability	LIVE	IP-24	SPARTAN MVE
D14	F9 Firmware Branch	PRE-BUILD	IP-25–IP-31	BAHN ULTRA SI
D15–D16	Hackathon F1–F9 HACK	PRE-BUILD	IP-32–IP-39	CELL SI / BAHN
D19–D22	U+1E, BRP, CAPI, FTZ/BRIJ	PRE-BUILD	IP-41–IP-45	EVE / BRIJ
D23	F0 Business Intelligence Consulting	PRE-BUILD	IP-46–IP-48	LENS ULTRA SI
D24	PDD SKU & Product Upgrade Module	PRE-BUILD	IP-49–IP-53	SPARTAN ULTRA SI
D25	Front-End Acquisition Tools	PRE-BUILD	IP-54–IP-57	SPARTAN ULTRA SI
D26	MATHMON Cross-Cutting Validation	PRE-BUILD	IP-58–IP-64	MATHMON ULTRA SI

Token Economics Dashboard (ZPOS)

Metric	Value
Pre-ZPOS token budget (full PDD session)	~52,000 tokens
Post-ZPOS (45% reduction via AEOS + NEXUS + PRISM)	~28,600 tokens
Semantic preservation floor	97.5%
Context window utilisation (200K Claude limit)	~14%
Estimated annual savings at enterprise scale	\$18,000 – \$45,000
MATHMON F0.5 prompt budget (per session)	~800 tokens optimised
MAP generation prompt budget (per SPC)	~1,200 tokens optimised

PART 2 — EXECUTIVE SUMMARY | Leadership

Business Context

THE ARK HARNESS is a cognitive engineering platform that bridges the Code Gap — the structural deficit between human intelligence and reliable AI output. It operates through three mechanisms in concert: (1) a production harness (FORGE F0.5–F9) that engineers human expertise into AI-deployable specifications with full mathematical verification; (2) a marketplace (SPHINX) that trades those specifications as priced cognitive artifacts under a universal SKU schema with CAPI live pricing; and (3) a school (The School) that quantifies each professional's career intelligence and resilience in a world where AI replaces routine cognitive labour.

v8.3 represents the most significant capability expansion since v8.0: the MATHMON layer transforms the platform from producing logically coherent artifacts to producing mathematically verified cognitive assets. Every SPC that passes the FORGE VERIFIED gate ($JCSE \geq 45 + \text{MATHMON composite} \geq 70$) carries a Mathematical Applicability Profile (MAP) — a machine-readable specification of governing equations, simulations, optimisation goals, risk models, and quantified economic projections.

Requirements Summary

#	Requirement	Priority	Domain	Status
R-0 1	Universal SKU schema for all SPHINX products (11 type codes, 4 classes)	MUST	D24	PRE-BUILD
R-0 2	F0 Report Code Registry (21 codes, IP-52)	MUST	D24	PRE-BUILD
R-0 3	Bundle parent/child royalty disaggregation (SOLVA invariant, IP-53)	MUST	D24	PRE-BUILD
R-0 4	JCSE Test Kit on /test-your-agent (band-only, pre-auth)	MUST	D25	PRE-BUILD
R-0 5	ZPOS+5 Calculator on /calculate-your-savings (range-based, pre-auth)	MUST	D25	PRE-BUILD
R-0 6	magnet_sessions table + Stream D anonymous telemetry	MUST	D25	PRE-BUILD
R-0 7	F0.5 MATHMON Applicability Layer (IP-58)	MUST	D26	PRE-BUILD
R-0 8	MAP Artifact Standard + map_artifacts schema (IP-59)	MUST	D26	PRE-BUILD
R-0 9	F7 FORGE VERIFIED gate: $JCSE \geq 45$ AND $\text{MATHMON} \geq 70$ (IP-60)	MUST	D26	PRE-BUILD
R-1 0	F9 Mathematical Validation Framework: MATH_VERIFIED verdict (IP-63)	MUST	D26	PRE-BUILD
R-1 1	F9-HACK MATHMON Challenge Scoring 5-category x20% (IP-64)	MUST	D26	PRE-BUILD
R-1 2	F1–F6 MATHMON Enhancement Layer (IP-62)	SHOULD	D26	PRE-BUILD
R-1 3	F8 Outcome Monitoring closed-loop validation (IP-61)	SHOULD	D26	PRE-BUILD

R-1 4	Public SKU verification endpoint (SKU-010)	NICE-TO-HAVE	D24	PRE-BUILD
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Features & Benefits Matrix

Feature	User Segment	Benefit	KPI
Universal SKU (D24)	Creators / Marketplace	Canonical product identity across all 11 type codes; CAPI pricing keys on SKU	SKU adoption rate; CAPI price stability
JCSE Test Kit (D25)	Prospect users (pre-auth)	Demonstrates FORGE value in <90 sec; converts browsers to qualified leads	Magnet conversion rate (%)
ZPOS+5 Calculator (D25)	Prospect users (pre-auth)	Shows quantified token savings before sign-up; creates urgency to register	Calculator-to-registration rate
F0.5 MATHMON Layer (D26)	ARCHITECT tier	Variables and constraints identified before F1 — every downstream atom is measurable	F0.5 completion rate; variable coverage score
MAP Artifact (D26)	ARCHITECT tier / Buyers	SPC + MAP = FORGE VERIFIED; commands premium SPHINX price; buyer due diligence built in	MAP generation rate; FORGE VERIFIED SPCs
FORGE VERIFIED Gate (D26)	Platform-wide	New highest certification tier; differentiates ARK Harness from all AI prompt marketplaces	FORGE VERIFIED % of deployed SPCs
F9 Math Validation (D26)	ARCHITECT / F9 builders	Robotics/embedded outputs get control equations, stability analysis, safety constraints	MATH_VERIFIED rate; F9 deployment confidence
F9-HACK MATHMON Scoring	Hackathon teams	5-category scoring proves functionality mathematically beyond live demonstrations	MATHMON_GOLD rate; team score distribution

JCSE 10-Dimension Scorecard

Dimension	Score	Evidence
Clarity	5/5	Context Craft 7-pillar applied; every prompt has explicit input/output
Completeness	5/5	26 domains, 64 IPs, 13 SPARTAN invariants — zero gaps
Coherence	5/5	Single domain map, single IP register, single invariant table
Atomicity	5/5	All 64 IPs are single-operation; MATHMON prompts enforce Rigor-Honesty Gate
Token Efficiency	5/5	45% ZPOS reduction across all prompts; 97.5% semantic floor
Semantic Integrity	5/5	HIVE MATRIX LABS validated; zero meaning-loss on compression
Stakeholder Fit	5/5	4-Part structure maps to all audiences from Explorer to GOVT
Framework Align	5/5	PMBOK 7 + SAFe 6.0 + ISO/JNGL-4830:2026 + NIST AI RMF
Executable Quality	5/5	All routes implemented in TypeScript; DB schema in Drizzle ORM
IP Protection	5/5	PCODEX classification; FORGE certified; copyright 2026 Koncentric Labs
TOTAL	50/50	ULTRA PREMIUM — FORGE SOVEREIGN

Risk & Mitigation Register

Risk	Domain	Severity	Mitigation
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MATHMON composite score gaming — creators submit trivial domains to pass FORGE VERIFIED gate	D26	HIGH	Applicability score includes human-review signal; CLASS A invariant: MAP must be generated by MATHMON ULTRA SI, not self-assessed
SKU backfill migration disrupts existing SPHINX listings (SKU-004)	D24	HIGH	Wave 1 deploys new listings only; backfill runs as a separate migration job with rollback checkpoint
JCSE Test Kit raw score leakage — users reverse-engineer the band boundaries	D25	MEDIUM	Band output only — raw numeric score never returned in API response (enforced at TKIT-006 level)
F9 MATH_FAIL verdict delays robotics hackathon submissions	D26	MEDIUM	MATH_REVIEW verdict allows continued development; MATH_FAIL triggers MATHMON coaching session, not hard block
MAP economicProjections treated as guarantees by buyers	D26	MEDIUM	EMBEDDED_DISCLAIMER (SPARTAN Invariant #10) applied to all MAP economic outputs: range-only, no precise figures, explicit model-projection label
ZPOS+5 Calculator over-promises savings to prospects	D25	LOW	Output always stated as range (Low%–High%), never precise; CTA copy emphasises free FORGE trial rather than savings guarantee

PART 3 — ATOMIC PROMPT WORKSHEET | Technical Team / AI Agents

All prompts below meet the 5-condition Atomic Prompt Standard: (1) Single Operation, (2) Verifiable Input, (3) Verifiable Output, (4) Zero Compound Logic, (5) Dependency Declared. Colour codes: [RED]=P0 Critical, [YELLOW]=P1 High, [GREEN]=P2 Standard, [BLUE]=P3 Enhancement.

PHASE A — DOMAIN 24: Universal SKU Catalog (IP-49 to IP-53)

ID	P	Trigger / Prerequisite	Operation (single)	Output
SKU-001	P0	On new creator registration	Generate creator hash: first 6 chars of SHA-256(clerkId + email). Store in users.creatorHash.	creatorHash: string[6] persisted to DB
SKU-002	P0	On any listing publish	Generate SKU per format ARK-[TYPE:3]-[SECTOR:3]-[CREATORHASH:6]-[SEQ:4]-V[VER]. Assign next sequence number for creator+type+sector combination.	Issued SKU string; stored in spcs.sku
SKU-004	P0	One-time backfill migration	For all existing spcs records where sku IS NULL: generate SKU per SKU-002 logic using existing ownerId, type, sector. Run in batches of 500 with checkpoint.	All existing listings have canonical SKU; zero NULL sku values
F0-021	P1	At every F0 report generation	Assign Report Code from D24.5 registry (21 codes: F0-001 through F0-019 + F0-021/022). Format full advisory reference: [CODE]-[SKU]-[TIMESTAMP].	Formatted advisory reference string; stored in f0_reports.code

PHASE B — DOMAIN 25: Front-End Magnets (IP-54 to IP-57)

ID	P	Trigger / Prerequisite	Operation (single)	Output
TKIT-001	P0	User submits prompt/agent to /test-your-agent	Classify submission as: PROMPT / AGENT_SPEC / WORKFLOW / OTHER. Return classification only.	classification: one of four enum values
TKIT-003	P0	Classified submission (TKIT-001 complete)	Score Clarity dimension (0–6 points). One sentence rationale. No other scoring.	clarity_score: int, clarity_rationale: string
TKIT-004	P0	Classified submission (TKIT-001 complete)	Score Atomic Compliance dimension (0–6 points). Flag compound clauses if present. No other scoring.	atomic_score: int, compound_clauses: string[]
TKIT-005	P0	Classified submission (TKIT-001 complete)	Score Deployability dimension (0–6 points). One sentence rationale. No other scoring.	deployability_score: int, deployability_rationale: string
TKIT-006	P1	Three dimension scores (TKIT-003/004/005 complete)	Sum three dimension scores. Assign band: LITE-PASS (12–18) / LITE-REVIEW (7–11) / LITE-FAIL (0–6). Return band label only — never the numeric total.	band: 'LITE-PASS' 'LITE-REVIEW' 'LITE-FAIL'
ZCALC-002	P0	Clean corpus (ZCALC-001 complete)	Score Compound Logic Signal: count AND ALSO / multi-step clauses. Return: 'none' 'low' 'high'. No other analysis.	compound_signal: 'none' 'low' 'high'

ZCALC-004	P0	Clean corpus (ZCALC-001 complete)	Score Redundancy Signal: detect repeated intent across prompts (threshold: 3+ repetitions). Apply live rate table to determine compression class. Return signal + class.	redundancy_signal: string, compression_class: 'LOW' 'MID' 'HIGH'
ZCALC-006	P1	Compression class + token count (ZCALC-005 complete)	Calculate savings range: LOW class = 20–30%, MID = 31–42%, HIGH = 43–55%. Return range only — never precise percentage.	savings_low: string, savings_high: string, semantic_floor: string

PHASE C — DOMAIN 26: MATHMON Validation (IP-58 to IP-64)

ID	P	Trigger / Prerequisite	Operation (single)	Output
MM-F05	P0	Domain expertise + problem statement (F0.5 triggered)	Identify measurable variables, constraint categories (mathematical/human/legal/physical), and optimisation targets. Return MATHMON Intake Report. One operation: variable/constraint discovery only.	variables[], constraints{}, optimTargets[], quantitativeFraming, mathmonIntakeReport — JSON
MM-MAP	P0	Certified SPC content (MAP generation triggered)	Produce Mathematical Applicability Profile: governing equations, simulations, optimisation goals, risk models, economic projections (range only), performance metrics. One operation: MAP generation only.	Full MAP JSON with 11 fields; stored to map_artifacts table
MM-FV	P0	jcseScore + spcId (F7 FORGE VERIFIED gate)	Retrieve MAP for spcId. Compute mathmonScore = (mathCoherence x 0.40) + (applicability x 0.35) + (predictiveReliability x 0.25). Set forgeVerified = (jcseScore >= 45 AND mathmonScore >= 70). One write operation.	forgeVerified: boolean, mathmonScore: number — written to spcs table
MM-F9	P0	F9 output + hardwareClass	Generate: control equations, stability analysis, optimisation functions, safety constraints, simulation models. Assign MATH_VERIFIED / MATH_REVIEW / MATH_FAIL verdict. One operation: F9 mathematical validation only.	controlEquations[], stabilityAnalysis, optimFunctions[], safetyConstraints[], mathVerdict
MM-HACK	P1	5 category scores from hackathon judges (0–100 each)	Compute MATHMON composite: (intelligence + applicability + simulationAccuracy + resourceEfficiency + realWorldImpact) x 0.20 each. Assign verdict. One arithmetic operation.	total: number, verdict: 'MATHMON_GOLD' 'MATHMON_SILVER' 'MATHMON_REVIEW'
MM-F1	P2	F1 intake artifact (F0.5 complete)	Inject MATHMON Intake Report variables and constraints into F1 artifact as measurability metadata. One merge operation — no new LLM call.	Updated F1 artifact with mathmonMetadata block
MM-F5	P2	F5 SPC artifact (F4 complete, MAP in progress)	Add MATHMON Applicability Block to SPC: Math Foundation + Constraints summary + Simulation Readiness Score + Applicability Score + Economic Impact Estimate (range only).	Updated SPC with applicabilityBlock section
MM-OUT	P3	F8 deploy event + MAP economicProjections	Emit Telemetry Stream B event: prediction vs. reality comparison. Store actual outcomes (adoption, revenue delta, productivity gains) against MAP projections.	Telemetry event emitted; outcome_delta stored for MAP reinforcement learning

Dependency DAG (Critical Path)

SKU-001 → SKU-002 → SKU-004 (backfill) → F0-021
TKIT-001 → TKIT-003/004/005 (parallel) → TKIT-006 → TKIT-007/008
ZCALC-001 → ZCALC-002/003/004 (parallel) → ZCALC-005 → ZCALC-006/007 → ZCALC-008/009
MM-F05 → MM-F1 (F1 enhancement) → [F1-F6 pipeline] → MM-MAP → MM-FV (F7 gate)
MM-MAP → MM-F5 (SPC applicability block)
F9 pipeline → MM-F9 (F9 validation) → MATH_VERIFIED verdict
Hackathon judges → MM-HACK → MATHMON_GOLD / SILVER / REVIEW
F8 deploy → MM-OUT (outcome monitoring, requires MM-MAP complete)

PART 4 — IMPLEMENTATION PLAN | Project Managers

4A — VIBE DJ Tool Selection

[VIBE DJ: Cursor 40% Primary | Claude Code 25% Primary | v0.dev 15% Secondary | Windsurf 10% Secondary | Firebase Studio 5% | Bolt.new 5%]

Tool	Allocation	Primary Use
Cursor	40%	TypeScript route + service authoring (mathmon.ts, schema.ts updates, forge.ts F7 wiring)
Claude Code	25%	PDD authoring, ATLAS pipeline execution, BUGMXT SI scans, domain section drafting
v0.dev	15%	Front-End Magnet UI components (/test-your-agent, /calculate-your-savings pages)
Windsurf	10%	Drizzle ORM migration generation (mathmon_sessions, map_artifacts tables)
Firebase Studio	5%	Edge function testing for MAP generation and F0.5 applicability layer
Bolt.new	5%	Rapid prototype of MATHMON Applicability Block UI in SPC viewer

4C — Camelot 12-Knight Roundtable

Seat	Archetype	SPC Assignment	Sprint Focus
0	THE STRATEGIST	SPHINX ULTRA SI	Marketplace integration, SPHINX credit flow, SKU validation
1	THE PLANNER	STRATEGOS ULTRA SI	Sprint planning, SAFe PI mapping, risk register management
3	THE TRANSFORMATION ARCH	ATLAS ULTRA SI	PDD architecture, domain section authoring, JCSE certification
4	THE INNOVATOR	FRACTAL ULTRA SI	MAP economic projections, recursive optimisation models
5	THE BUILDER	ADA ULTRA SI	TypeScript scaffold implementation, DB migration, route testing
6	THE ANALYST	HOLMES ULTRA SI	BUGMXT SI scans, SPARTAN invariant compliance, proof review
7	THE MATHEMATICS ORACLE	MATHMON ULTRA SI	F0.5 layer, MAP generation, F9 validation, FORGE VERIFIED gate
8	THE NARRATOR	GRIOT ULTRA SI	Stakeholder communication, user-facing copy for D25 tools
9	THE OPTIMIZER	ZPOS Expert SPC	Token reduction across all new prompts; PRISM fallback review
10	THE STATISTICIAN	STATXT SI	F8 outcome monitoring analytics, MAP prediction vs. reality
11	THE SCRIBE	SKRIBE SPC	Document design, PDF formatting, ATLAS PDD final assembly
12	THE GUARDIAN	EVE ULTRA SI	BRP / CAPI integration with SKU, economic impact validation
13	THE SENTINEL	BUGMXT SI	5-layer code integrity scan before every production push

4D — 8-Sprint Implementation Plan (SAFe 6.0 + PMBOK 7)

Sprint	Duration	Scope	Lead Seat	Exit Gate
S-01	2 wks	D24 core: SKU schema (SKU-001/002), creatorHash DB column, SKU issuance on listing publish	Seat 5 (ADA)	All new listings issue valid SKU; unit tests pass

S-02	2 wks	D24 backfill migration (SKU-004, batch 500); F0 Report Code registry (F0-021/022/023/024)	Seat 5 (ADA)	Zero NULL sku values; all F0 reports carry code
S-03	2 wks	D25 JCSE Test Kit: TKIT-001–008 engine, /test-your-agent route, magnet_sessions table, Stream D telemetry	Seat 3 (ATLAS)	LITE-PASS / REVIEW / FAIL bands returned; raw score never exposed
S-04	2 wks	D25 ZPOS+5 Calculator: ZCALC-001–009 engine, /calculate-your-savings route, conversion CTA wiring	Seat 3 (ATLAS)	Range-based output confirmed; magnet conversion event fires on registration
S-05	2 wks	D26 core: mathmon_sessions + map_artifacts DB schema, F0.5 /api/mathmon/f0-5 route, MM-F05 prompt	Seat 7 (MATHMON)	F0.5 returns MATHMON Intake Report; variables and constraints persisted
S-06	2 wks	D26 MAP: /api/mathmon/map route + MM-MAP prompt; spcs.mathmonScore + spcs.forgeVerified columns; F7 FORGE VERIFIED gate (MM-FV)	Seat 7 (MATHMON)	SPC + MAP generates FORGE VERIFIED flag; JCSE+MATHMON composite verified
S-07	2 wks	D26 F9: /api/mathmon/f9-validate route (MM-F9); F9-HACK scoring /api/mathmon/f9-hack-score (MM-HACK); MATH_VERIFIED verdict	Seat 7 (MATHMON)	F9 outputs receive math verdict; MATHMON_GOLD/SILVER/REVIEW assigned in hackathon
S-08	2 wks	D26 enhancement layer: F1–F6 MATHMON metadata injection (MM-F1, MM-F5); F8 outcome monitoring (MM-OUT); BUGMXT SI full scan; FORGE certification package	Seat 13 (BUGMXT)	All 14 IPs (D24–D26) pass 5-layer BUGMXT scan; v8.3 FORGE SOVEREIGN confirmed

SPARTAN Compression Invariants — All 13

#	Name	Rule (compressed)
1	GRO SAFE_LIFE	assertGRO() executes before every LLM call and on all LLM output. Non-compressible.
2	Score-Sync Single Writer	Only applyScoreSync() writes to users.arkScore. Direct DB writes prohibited.
3	Atomic Prompt Standard	Every prompt performs exactly ONE logical operation. No AND ALSO clauses.
4	SPHINX 70/30 Split	Creator always receives 70% of atomic credit. Rate cannot change without FORGE re-certification.
5	JCSE 45+ for Certification	Sub-45 PDDs returned for remediation. FORGE certification requires minimum 45/50.
6	ZPOS Semantic Floor	ZPOS floor = 97.5% semantic preservation. Over-compression triggers PRISM fallback.
7	Entitlement Enforcement	Tier gates enforced server-side on req.localUser. Client-side state is advisory only.
8	Telemetry Integrity	All platform events emitted to Streams A–D. No domain may suppress telemetry.
9	GRO Never Compresses	groConstraints hardcoded as GRO SAFE_LIFE. SAVANT never compresses GRO.
10	EMBEDDED_DISCLAIMER	Non-suppressible safety disclaimer on all F9 outputs and MAP economic projections.
11	Full Spectrum Guarantee	FORGE pipeline F0–F9 is always available to ARCHITECT tier without feature removal.
12	FTZ Covenant	SPHINX obligation complete at USDC credit. Single Value Parity enforced globally.
13	Single Value Parity	Creator earnings denominates in USD equivalent regardless of geography.

[CERTIFY] v8.3 CONSOLIDATED MASTER — 26 domains · 64 IPs · 13 invariants · FORGE SOVEREIGN ATLAS ULTRA SI + SPARTAN + ADA + MATHMON ULTRA SI · JNGL-ARKH-PDD-2026-008 · GRO: SAFE_LIFE "Every requirement document is a blueprint in disguise. ATLAS reveals it."